STAT 559 – Note 03
Measurable Functions and Lebesgue Integration

Edited transcription of handwritten lecture notes

These notes transcribe and edit the indicated handwritten PDF. Source-page markers preserve traceability. Yellow boxes retain original TODOs; red boxes explain mathematical corrections; blue boxes add explanations or proof details. The reference is *stats310notes.pdf* (provided in `notes/`); references below use its printed section numbers.

Source page(s): 1

1 Motivation

Lebesgue integration starts with a measure and builds integration from it. On a general space we write $\int_\Omega f(\omega)\, d\mu(\omega)$. With Lebesgue measure on a real interval this agrees with the Riemann integral whenever the bounded function is Riemann integrable (reference Exercise 2.4.2). It applies to many more measurable functions and supports powerful limit theorems.

Correction: Existence of an integral

Measurability alone does not guarantee a finite integral, or even a defined signed integral. A nonnegative measurable function always has an extended integral in $[0, \infty]$. A signed integral must avoid the indeterminate expression infinity minus infinity.

Source page(s): 2–3

2 Measurable functions

For measurable spaces $(\Omega, \mathcal{F})$ and $(\Omega', \mathcal{F}')$, a map $f : \Omega \to \Omega'$ is measurable if $f^{-1}(A) \in \mathcal{F}$ for every $A \in \mathcal{F}'$. **Generator criterion** (reference Lemma 2.1.3). If $\mathcal{F}' = \sigma(\mathcal{A})$ and $f^{-1}(A) \in \mathcal{F}$ for all $A \in \mathcal{A}$, then f is measurable. Indeed,

$$\mathcal{G} = \{B \subseteq \Omega' : f^{-1}(B) \in \mathcal{F}\} \tag{1}$$

is a sigma-algebra because inverse images preserve complements and countable unions. It contains $\mathcal{A}$, hence $\mathcal{F}'$. Continuous maps between topological spaces are Borel measurable: the inverse image of an open set is open, and open sets generate the Borel sigma-algebra. Compositions of measurable maps are measurable, since $(g \circ f)^{-1}(A) = f^{-1}(g^{-1}(A))$. For real-valued measurable f, g, the functions $f + g, f - g, fg$ are measurable. The quotient f/g is measurable wherever $g \neq 0$, with a specified measurable extension on its zero set if needed.

Source page(s): 4–5

3 Extended real functions and limits

Write $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, \infty\}$, with its order topology and Borel sigma-algebra. Countable infima and suprema of measurable extended real functions are measurable. For example,

$$\left\{\inf_n f_n \geq t\right\} = \bigcap_n \{f_n \geq t\}. \tag{2}$$

The upper rays $[t, \infty]$ for finite real t generate $\mathcal{B}(\overline{\mathbb{R}})$. The supremum result follows by negation. Thus

$$\liminf_n f_n = \sup_k \inf_{n \geq k} f_n, \tag{3}$$

$$\limsup_n f_n = \inf_k \sup_{n \geq k} f_n \tag{4}$$

are measurable, as is any everywhere pointwise limit.

Supplement: The source's question about open sets

The open sets in $\mathbb{R}$ and $\overline{\mathbb{R}}$ are not literally the same: neighborhoods of the added endpoints include sets such as $(a, \infty]$. On $\mathbb{R}$ an open set is a countable union of open intervals, but an arbitrary open set need not be a single interval. Its usual topology is the subspace topology inherited from $\overline{\mathbb{R}}$.

Source page(s): 6–8

4 Four stages in defining the integral

Fix a measure space $(\Omega, \mathcal{F}, \mu)$. Measurability of f refers to the domain sigma-algebra and target Borel sigma-algebra; the domain measure supplies the integral. First, for $A \in \mathcal{F}$, set

$$\int \mathbf{1}_A \, d\mu = \mu(A). \tag{5}$$

Second, a *nonnegative simple function* has a finite representation $s = \sum_{i=1}^m a_i \mathbf{1}_{A_i}$ with finite $a_i \geq 0$ and disjoint measurable A_i. Define

$$\int s \, d\mu = \sum_{i=1}^m a_i \mu(A_i), \qquad 0 \cdot \infty := 0. \tag{6}$$

Third, for measurable $f \geq 0$, let $\mathcal{S}_f$ be the nonnegative simple functions bounded above by f. Define

$$\int f \, d\mu = \sup_{s \in \mathcal{S}_f} \int s \, d\mu. \tag{7}$$

Original TODO: Prove linearity for simple functions

Refine two finite simple representations by the measurable intersections of their level sets (including zero level sets to partition the domain). On each common cell both functions are constant. Distribute the finite sums to prove

$$\int (as + bt)\, d\mu = a \int s\, d\mu + b \int t\, d\mu \quad (a, b \geq 0). \tag{8}$$

The same refinement proves that the integral is independent of the chosen representation. See reference Section 2.2.

Source page(s): 9–10

Fourth, put $f^+ = \max(f, 0)$ and $f^- = \max(-f, 0)$. Then $f = f^+ - f^-$ and $|f| = f^+ + f^-$. Define

$$\int f\, d\mu = \int f^+\, d\mu - \int f^-\, d\mu \tag{9}$$

if at least one term is finite. The function is *integrable* when both are finite, equivalently $\int |f|\, d\mu < \infty$. For measurable S, $\int_S f\, d\mu = \int f \mathbf{1}_S\, d\mu$, interpreting the product to be zero outside S even if f is infinite there.

Original TODO: An undefined signed integral

On $\mathbb{R}$ with Lebesgue measure, let $f(x) = 1$ for $x \geq 0$ and $f(x) = -1$ for $x < 0$. Both $\int f^+\, d\lambda$ and $\int f^-\, d\lambda$ are infinite, so its signed Lebesgue integral is undefined.

Source page(s): 11–12

5 Monotone convergence and simple approximation

Monotone convergence theorem (MCT). For measurable $0 \leq f_n \uparrow f$ pointwise,

$$\lim_n \int f_n\, d\mu = \int f\, d\mu. \tag{10}$$

The limit f is measurable. A key companion fact is that every measurable $f \geq 0$ is an increasing limit of nonnegative simple functions (reference Proposition 2.3.6).

Supplement: The construction assigned as homework

One explicit approximation is

$$s_n = 2^{-n} \lfloor 2^n \min(f, n) \rfloor . \tag{11}$$

For $f = \infty$ take $\min(f, n) = n$. Each s_n has finitely many measurable level sets; refinement of the dyadic grid and increasing truncation show $s_n \leq s_{n+1} \leq f$. For finite $f(\omega)$, the approximation error is eventually less than 2^{-n}; at infinity, $s_n(\omega) = n \to \infty$. Thus $s_n \uparrow f$.

Source page(s): 13–14

6 Linearity

For measurable nonnegative f, g, choose $u_n \uparrow f$ and $v_n \uparrow g$ simple. For $a, b \geq 0$, MCT and simple-function linearity give

$$\int (af + bg)\, d\mu = \lim_n \int (au_n + bv_n)\, d\mu \tag{12}$$

$$= a \lim_n \int u_n\, d\mu + b \lim_n \int v_n\, d\mu \tag{13}$$

$$= a \int f\, d\mu + b \int g\, d\mu. \tag{14}$$

Original TODO: Complete the signed linearity proof

For integrable f, g, their infinite values occur only on a measurable null set. Define ambiguous sums there to be zero. The inequalities $(f+g)^{\pm} \leq f^{\pm} + g^{\pm}$ show integrability. The identity

$$(f+g)^+ + f^- + g^- = (f+g)^- + f^+ + g^+ \tag{15}$$

holds off that null set, and the exceptional set contributes zero. Integrate using nonnegative additivity and rearrange finite numbers to obtain $\int (f+g)\, d\mu = \int f\, d\mu + \int g\, d\mu$. For a negative scalar, positive and negative parts exchange roles. Consequently all real a, b satisfy

$$\int (af + bg)\, d\mu = a \int f\, d\mu + b \int g\, d\mu. \tag{16}$$

See reference Proposition 2.4.1.

Correction: Undefined pointwise sums

The source writes $\infty + (-\infty) = 0$. This is not an arithmetic identity. For integrable functions one may *choose* zero on the null set where the sum is otherwise undefined, without affecting any integral.

Source page(s): 15–17

7 Proof of MCT: preparatory facts

If $0 \leq f \leq g$, then $\mathcal{S}_f \subseteq \mathcal{S}_g$, so $\int f\, d\mu \leq \int g\, d\mu$. For a fixed nonnegative simple function $s = \sum_{i=1}^m a_i \mathbf{1}_{A_i}$ define a *set function*

$$\nu(S) = \int_S s\, d\mu = \sum_{i=1}^m a_i \mu(A_i \cap S), \qquad S \in \mathcal{F}. \tag{17}$$

It is a measure: nonnegativity and the empty-set condition are immediate. For disjoint S_j,

$$\nu\left(\bigcup_j S_j\right) = \sum_{i=1}^{m} a_i \sum_j \mu(A_i \cap S_j) \tag{18}$$

$$= \sum_j \sum_{i=1}^{m} a_i \mu(A_i \cap S_j) = \sum_j \nu(S_j). \tag{19}$$

Here all terms are nonnegative. This answers the source's question: S is a measurable *set*; s is the simple *function*.

Source page(s): 18–19

8 Proof of MCT: main step

Monotonicity gives $L := \lim_n \int f_n \, d\mu \le \int f \, d\mu$. Fix $s \in \mathcal{S}_f$ and $0 < \alpha < 1$, and set $S_n = \{\alpha s \le f_n\}$. Then $S_n \uparrow \Omega$: where $s = 0$ the condition holds immediately; where $s > 0$, convergence to $f \ge s > \alpha s$ makes it eventually hold. Continuity from below of the measure ν gives

$$\alpha \int s \, d\mu = \lim_n \int_{S_n} \alpha s \, d\mu \le \lim_n \int f_n \, d\mu = L. \tag{20}$$

Take the supremum over s and let $\alpha \uparrow 1$ to obtain the reverse inequality.

Original TODO: Why choose alpha strictly below one?

For $f_n = 1 - 1/(n+1)$ and $s = f = 1$, the sets $\{s \le f_n\}$ are empty for every n. Choosing $\alpha < 1$ gives a strict gap and guarantees eventual inclusion. This is the reason for the extra parameter in reference Theorem 2.3.2.

Source page(s): 20–21

9 Fatou's lemma

For measurable $f_n \ge 0$,

$$\int \liminf_n f_n \, d\mu \le \liminf_n \int f_n \, d\mu. \tag{21}$$

Set $g_k = \inf_{n \ge k} f_n$. Then $g_k \uparrow \liminf_n f_n$ and, for every $n \ge k$, $g_k \le f_n$. Therefore

$$\int g_k \, d\mu \le \inf_{n \ge k} \int f_n \, d\mu. \tag{22}$$

MCT on the left and the definition of liminf on the right prove the assertion (reference Theorem 2.5.1).

Original TODO: Midterm: redo Fatou's proof

The source flags this proof for revision. The essential move is to replace the original sequence by its increasing tail infima, then use MCT. Nonnegativity is the hypothesis that permits MCT.

Correction: Limits cannot be assumed to exist

The handwritten chain uses $\lim_n \int f_n \, d\mu$ before establishing its existence, and ends with "LHS = RHS." Fatou gives an inequality, not generally equality. The tail-infimum comparison above is the valid argument. For strictness, on $\mathbb{R}$ let $f_n = \mathbf{1}_{[n,n+1]}$; then the integral of the liminf is zero and every integral is one.

Source page(s): 22–24

10 Dominated convergence theorem (DCT)

Suppose measurable $f_n \to f$ pointwise and $|f_n| \le h$ for some nonnegative integrable h. Then f and every f_n are integrable, and

$$\lim_n \int f_n \, d\mu = \int f \, d\mu, \tag{23}$$

$$\lim_n \int |f_n - f| \, d\mu = 0. \tag{24}$$

Proof. We have $|f| \le h$. Apply Fatou to $f_n + h \ge 0$ and subtract the finite number $\int h \, d\mu$:

$$\int f \, d\mu \le \liminf_n \int f_n \, d\mu. \tag{25}$$

Apply the same argument to $-f_n$ to obtain

$$\int f \, d\mu \ge \limsup_n \int f_n \, d\mu. \tag{26}$$

The two bounds prove convergence of the integrals. Apply this result to $|f_n - f| \to 0$, dominated by $2h$, to prove the second assertion (reference Theorem 2.5.2).

Original TODO: Why does convergence in integral absolute difference imply convergence of integrals?

Use

$$\left| \int f_n \, d\mu - \int f \, d\mu \right| \le \int |f_n - f| \, d\mu. \tag{27}$$

The right side tends to zero. This implication does not itself require the full DCT assumptions once integrability and that convergence are given.

Correction: The dominating function must be absolutely integrable

The source sometimes writes only $\int h \, d\mu < \infty$ while placing $|h|$ in the bound. The correct requirement is an integrable nonnegative bound, or equivalently $\int |h| \, d\mu < \infty$. A negative infinite signed integral is not a usable bound.

Source page(s): 25–28

11 Counterexamples and differentiation under the integral

MCT and Fatou in their stated forms require nonnegativity; DCT allows signs.

Original TODO: A counterexample without domination

On $(0,1)$ with Lebesgue measure let $f_n(x) = n\mathbf{1}_{(0,1/n)}(x)$. Then $f_n \to 0$ pointwise and $\int f_n \, d\lambda = 1$. Thus pointwise convergence alone does not allow interchanging a limit and an integral. Domination is a sufficient hypothesis for DCT; it is not a necessary condition for every particular sequence whose integrals happen to converge.

The statistical motivation is a likelihood $\ell_n(\theta) = \sum_{i=1}^n \log p_\theta(X_i)$ for iid data from p_{θ_0}. Under suitable regularity, the score has mean zero:

$$\mathbb{E}_{\theta_0}\left[\frac{\partial}{\partial\theta}\log p_\theta(X)\Big|_{\theta=\theta_0}\right] = \int \frac{\partial p_\theta(x)}{\partial\theta}\,dx\Big|_{\theta=\theta_0} = \frac{\partial}{\partial\theta}\int p_\theta(x)\,dx\Big|_{\theta=\theta_0} = 0. \tag{28}$$

Correction: Regularity for the score argument

The exchange of derivative and integral needs justification, including a common dominating measure, differentiability, and local integrable domination of derivatives. The logarithmic expression also needs positivity almost surely on the relevant support. A zero expected score and a law of large numbers alone do not prove consistency of an arbitrary maximum likelihood estimator; further identification and maximization arguments are needed.

Leibniz rule (reference Proposition 2.5.5). Let $I \subseteq \mathbb{R}$ be open and $F : I \times \Omega \to \mathbb{R}$. Assume each $F(x,\cdot)$ is integrable, each $F(\cdot,\omega)$ is differentiable, and $|F_x(x,\omega)| \le h(\omega)$ for an integrable nonnegative h. Then

$$\frac{d}{dx}\int F(x,\omega)\,d\mu(\omega) = \int F_x(x,\omega)\,d\mu(\omega). \tag{29}$$

Fix x and $x_n \to x$ with $x_n \neq x$ and eventually in an interval contained in I. The measurable difference quotients

$$g_n(\omega) = \frac{F(x,\omega) - F(x_n,\omega)}{x - x_n} \tag{30}$$

converge to $F_x(x,\omega)$. The mean value theorem gives $|g_n| \le h$. DCT and linearity identify their integral limit with the derivative of the integral. Since the argument works for every such sequence, the derivative exists as asserted.

Source page(s): 29–30

12 Almost everywhere and almost surely

A property holds *almost everywhere* (a.e.) if it holds outside a measurable null set. In a probability space the same notion is called *almost surely* (a.s.). In particular $f = g$ a.e. means $\{f \neq g\}$ is contained in a measurable null set. If f, g are measurable extended real functions, $\{f \neq g\}$ is measurable, so this is equivalent to $\mu(\{f \neq g\}) = 0$. Without measurability the exceptional set need not itself be measurable. In a complete space every subset of a null set is measurable.

Original TODO: Nonmeasurable functions that agree almost everywhere; what is the difference?

Use the incomplete finite space from Note 02: $\Omega = \{a, b, c\}$ with measurable atoms $\{a, b\}$ (mass zero) and $\{c\}$ (mass one). Let $f = \mathbf{1}_{\{a\}}$ and $g = \mathbf{1}_{\{b\}}$. Both are nonmeasurable, yet $f = g = 0$ outside the measurable null set $\{a, b\}$. Also f agrees a.e. with the measurable zero function, while $\{f \neq 0\} = \{a\}$ is not measurable. This explains why containment in a measurable null set is the general definition. The source references an earlier lecture date here, without specifying another example.

Similarly $f_n \to f$ a.e. means convergence outside one measurable null set, even when measurability of the individual functions has not been assumed (reference Section 2.6).

Source page(s): 30–32

13 Zero integral of a nonnegative function

For measurable $f \geq 0$,

$$\int f \, d\mu = 0 \quad \Longleftrightarrow \quad f = 0 \text{ a.e.} \tag{31}$$

If $f = 0$ a.e., every simple $0 \leq s \leq f$ is supported on a null set and has zero integral. Taking the supremum gives the forward integral assertion. Conversely, if $\mu(\{f > 0\}) > 0$, then

$$\{f > 0\} = \bigcup_{n=1}^{\infty} \{f > 1/n\} \tag{32}$$

implies that some $A_n = \{f > 1/n\}$ has positive measure. Hence

$$\int f \, d\mu \geq \int n^{-1} \mathbf{1}_{A_n} \, d\mu = n^{-1} \mu(A_n) > 0, \tag{33}$$

a contradiction.

Original TODO: Why may an integrable function be infinite somewhere?

Integrability forces its infinite-value set to be null, not empty. For example, $f = \infty \mathbf{1}_{\{0\}}$ on $\mathbb{R}$, interpreted as zero away from zero, is measurable and has integral zero. The convention $0 \cdot \infty = 0$ in integration captures the zero measure of its support. This also explains the source's final "think why" note.